

GIT and GITHUB

1. Create a GitHub account

Step 1: Open GitHub

Go to: [GitHub](https://github.com)

Step 2: Click Sign up

Click the Signup button in the top-right corner.

Step 3: Enter Your Email

Enter your email address and click Continue.

Step 4: Create a Password

Create a strong password and click Continue.

Step 5: Choose a Username

Ex: rajkumari

Step 6: Complete the Verification

Step 7: Verify Your Email

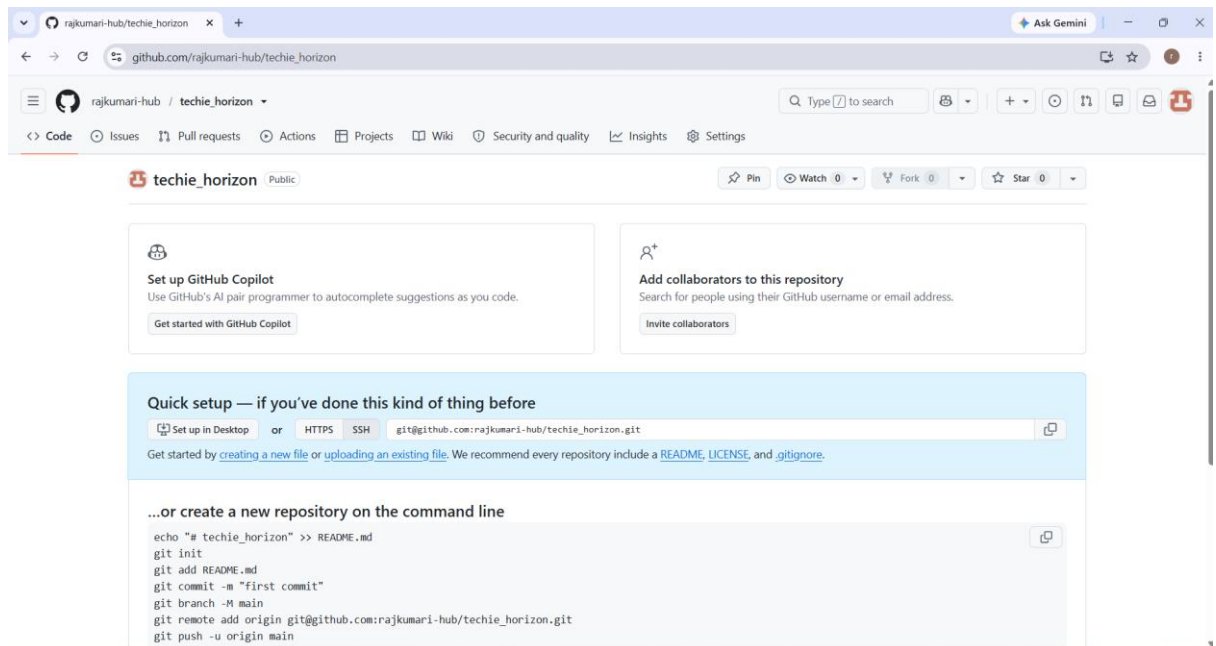
Step 10: Create Your First Repository

After logging in:

Click the + icon (top-right).

Select New repository.

Repository name: techie_horizon



2. To SetUp MFA (Multi-Factor Authentication) for GitHub Using a Mobile App

Step 1: Sign in to GitHub

Go to:

[GitHub](#)

Log in to your account.

Step 2: Open Settings

Click your profile picture (top-right).

Click Settings.

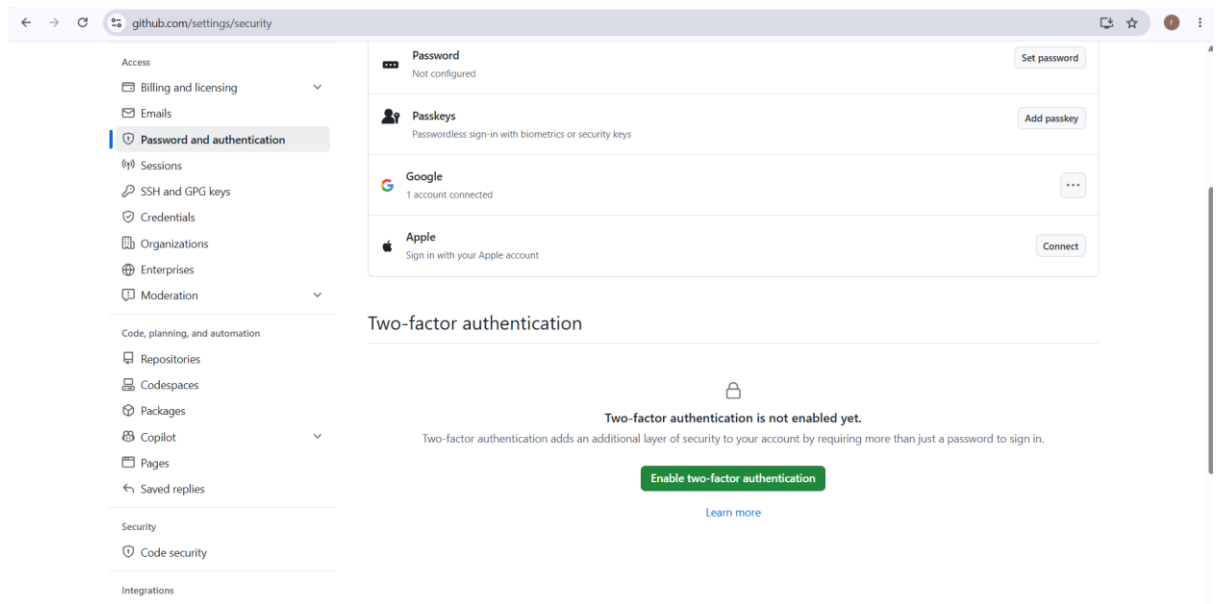
Step 3: Open Password and Authentication

In the left menu, click: Password and authentication

Step 4: Start MFA Setup

Scroll to Two-factor authentication.

Click: Enable two-factor authentication



Step 5: Choose Authentication App

Select:

Authentication App

Step 6: Install an Authenticator App on Your Phone

Install any one of these:

Google Authenticator

Step 7: Scan the QR Code

GitHub will display a QR code.

On your phone:

Tap + (Add Account).

Choose Scan QR Code.

Scan the QR code shown on GitHub.

Setup authenticator app

Authenticator apps and browser extensions like [1Password](#), [Authy](#), [Microsoft Authenticator](#), etc. generate one-time passwords that are used as a second factor to verify your identity when prompted during sign-in.

Scan the QR code

Use an authenticator app or browser extension to scan. [Learn more about enabling 2FA.](#)



Unable to scan? You can use the [setup key](#) to manually configure your authenticator app.

Verify the code from the app

[Cancel](#) [Continue](#)

Alternative 2FA option:

Step 8: Enter the Verification Code

The app will generate a 6-digit code.

Enter that code into GitHub and click:

Step 9: Verify MFA

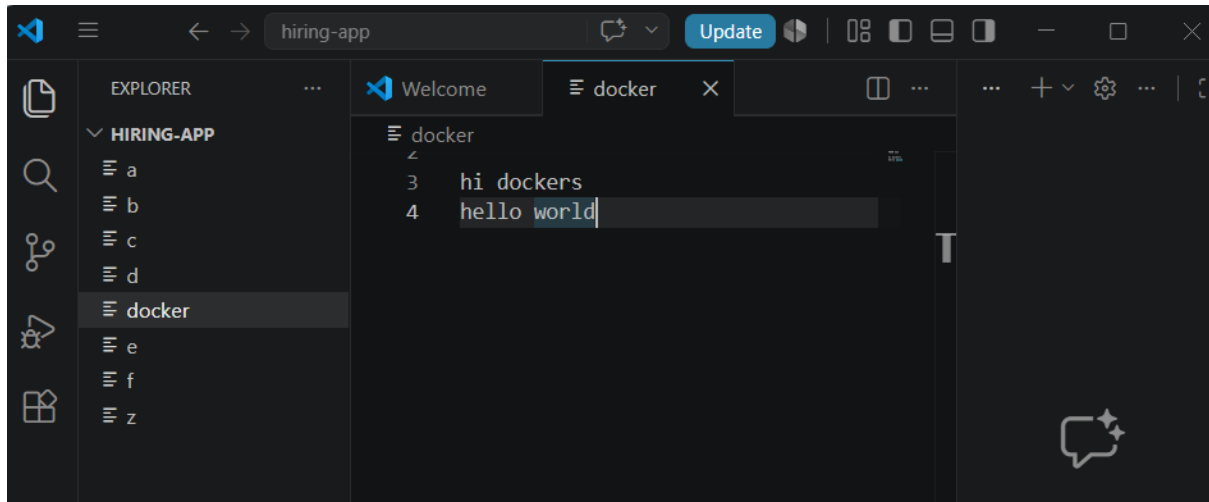
1. Sign out of GitHub.
2. Sign in again.
3. Enter:
 - Your password.
 - The 6-digit code from your authenticator app

3. Execute all commands shown in the video

Create directory

\$ mkdir rajkumari-techie or Using VS code

\$ cd rajkumari-techie



```
brajk@RajKumari MINGW64 ~/OneDrive/Desktop
$ cd

brajk@RajKumari MINGW64 ~
$ mkdir rajkumari-techie

brajk@RajKumari MINGW64 ~
$ cd rajkumari-techie

brajk@RajKumari MINGW64 ~/rajkumari-techie
$ |
```

Add User, Email and Check

\$ git config --global user.name rajkumari

\$ git config --global user.email brajkumari7675@gmail.com

\$ git config --list

```

brajk@RajKumari MINGW64 ~/rajkumari-techie
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager-core
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=rajkumari
user.email=brajkumari7675@gmail.com

brajk@RajKumari MINGW64 ~/rajkumari-techie
$ |

```

Initialise git

\$ git init

Create files

\$ touch file1.txt file2.txt file3.txt

Add files to staging

\$ git add file1.txt file2.txt file3.txt

Check status

\$ git status

```

brajk@RajKumari MINGW64 ~/rajkumari-techie (master)
$ touch file1.txt file2.txt file3.txt

brajk@RajKumari MINGW64 ~/rajkumari-techie (master)
$ git add file1.txt file2.txt file3.txt

brajk@RajKumari MINGW64 ~/rajkumari-techie (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   file1.txt
    new file:   file2.txt
    new file:   file3.txt

brajk@RajKumari MINGW64 ~/rajkumari-techie (master)
$ |

```

Add repository by using url of repo

```
$ git remote add origin git@github.com:rajkumari-hub/techie_horizon.git
```

Confirm the repository connection

```
$ git remote -v
```

Commit the file

```
$ git commit -m "1st commit"
```

Check commit logs

```
$ git log -4
```

```
brajk@RajKumari MINGW64 ~/rajkumari-techie (master)
$ git remote -v
origin  git@github.com:rajkumari-hub/techie_horizon.git (fetch)
origin  git@github.com:rajkumari-hub/techie_horizon.git (push)

brajk@RajKumari MINGW64 ~/rajkumari-techie (master)
$ git commit -m "file commit"
[master (root-commit) e9dcc0d] file commit
3 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 file1.txt
create mode 100644 file2.txt
create mode 100644 file3.txt

brajk@RajKumari MINGW64 ~/rajkumari-techie (master)
$ git log -4
commit e9dcc0dc3cccdf330aaf5126d162bcc3fc6aab07 (HEAD -> master)
Author: rajkumari <brajkumari7675@gmail.com>
Date:   Tue Jul 7 18:02:28 2026 +0530

    file commit

brajk@RajKumari MINGW64 ~/rajkumari-techie (master)
$
```

Push your files to Github Repostitory

```
$ git push -u origin master
```

```

brajk@RajKumari MINGW64 ~/rajkumari-techie (master)
$ ls -la ~/.ssh
total 44
drwxr-xr-x 1 brajk 197609  0 Jul  2 11:51 ./
drwxr-xr-x 1 brajk 197609  0 Jul  7 18:05 ../
-rw-r--r-- 1 brajk 197609 3381 Jun 28 15:07 id_rsa
-rw-r--r-- 1 brajk 197609  741 Jun 28 15:07 id_rsa.pub
-rw-r--r-- 1 brajk 197609 1993 Jul  7 18:07 known_hosts
-rw-r--r-- 1 brajk 197609 1442 Jun 30 23:49 known_hosts.old

brajk@RajKumari MINGW64 ~/rajkumari-techie (master)
$ git remote -v
origin  git@github.com:rajkumari-hub/techie_horizon.git (fetch)
origin  git@github.com:rajkumari-hub/techie_horizon.git (push)

brajk@RajKumari MINGW64 ~/rajkumari-techie (master)
$ |

```

```

brajk@RajKumari MINGW64 ~/rajkumari-techie (master)
$ git push -u origin master
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 219 bytes | 219.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:rajkumari-hub/techie_horizon.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

brajk@RajKumari MINGW64 ~/rajkumari-techie (master)
$ |

```

The screenshot shows a web browser displaying the GitHub repository page for 'techie_horizon' by user 'rajkumari-hub'. The repository is public and has 0 stars, 0 forks, and 0 watchers. The main content area shows a file commit table with three files: 'file1.txt', 'file2.txt', and 'file3.txt', all committed 27 minutes ago. Below the table is a 'README' section with a placeholder for a README file, including a button to 'Add a README'. The right sidebar contains sections for 'About' (no description provided), 'Releases' (no releases published), 'Packages' (no packages published), and 'Contributors' (no contributors).

Git Branching Steps

Create new branch

```
$ git branch feature
```

Change the branch

```
$ git checkout feature
```

Delete branch

```
$ git branch -d master
```

Check list of branches

```
$ git branch
```

```
prajk@RajKumari MINGW64 ~/rajkumari-techie (master)
$ git branch feature

prajk@RajKumari MINGW64 ~/rajkumari-techie (master)
$ git checkout feature
Switched to branch 'feature'

prajk@RajKumari MINGW64 ~/rajkumari-techie (feature)
$ git branch -d main
error: branch 'main' not found.

prajk@RajKumari MINGW64 ~/rajkumari-techie (feature)
$ git branch -d master
Deleted branch master (was e9dcc0d).

prajk@RajKumari MINGW64 ~/rajkumari-techie (feature)
$ git switch main
fatal: invalid reference: main

prajk@RajKumari MINGW64 ~/rajkumari-techie (feature)
$ git branch main

prajk@RajKumari MINGW64 ~/rajkumari-techie (feature)
$ git switch main
Switched to branch 'main'

prajk@RajKumari MINGW64 ~/rajkumari-techie (main)
$ git branch
  feature
* main

prajk@RajKumari MINGW64 ~/rajkumari-techie (main)
$ |
```

Pulling from Github

\$ git pull

```
brajk@RajKumari MINGW64 ~/rajkumari-techie (main)
$ git pull
There is no tracking information for the current branch.
Please specify which branch you want to merge with.
See git-pull(1) for details.

    git pull <remote> <branch>

If you wish to set tracking information for this branch you can do so with:

    git branch --set-upstream-to=origin/<branch> main

brajk@RajKumari MINGW64 ~/rajkumari-techie (main)
$ git switch master
Switched to a new branch 'master'
branch 'master' set up to track 'origin/master'.

brajk@RajKumari MINGW64 ~/rajkumari-techie (master)
$ git pull
Already up to date.

brajk@RajKumari MINGW64 ~/rajkumari-techie (master)
$ |
```

To get clone copy

\$ git clone https://github.com/rajkumari-hub/techie_horizon.git

```
brajk@RajKumari MINGW64 ~/rajkumari-techie (master)
$ git clone https://github.com/rajkumari-hub/techie_horizon.git
Cloning into 'techie_horizon'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 3 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

brajk@RajKumari MINGW64 ~/rajkumari-techie (master)
$ ls
file1.txt file2.txt file3.txt techie_horizon/

brajk@RajKumari MINGW64 ~/rajkumari-techie (master)
$ |
```

